

THE UNIVERSITY OF MELBOURNE

ELEN90098

Reinforcement Learning for Engineering

Workshop 1

Multi-Armed Bandits

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Question 1

Let $\mathcal{A}$ denote the set of available arms. At time step k , the algorithm selects an arm $A_k \in \mathcal{A}$ and receives a random reward R_k . The true action value of arm a is

$$q^*(a) = \mathbb{E}[R_k \mid A_k = a],$$

which is the expected reward obtained when arm a is selected.

Let

$$a^* \in \arg \max_{a \in \mathcal{A}} q^*(a)$$

be an optimal arm. The best achievable expected reward per time step is then

$$R^* = q^*(a^*) = \max_{a \in \mathcal{A}} q^*(a).$$

Thus, R^* is a deterministic expected reward, whereas R_k is the random reward actually observed at time step k .

The cumulative expected regret after t time steps is defined as

$$\text{Regret}_t = \mathbb{E} \left[\sum_{k=1}^t (R^* - R_k) \right]. \quad (1.1)$$

The expectation in (1.1) accounts for the randomness in the observed rewards and, where applicable, the randomisation of the bandit policy. In particular, if arm a is selected at time k , its expected one-step regret is

$$R^* - q^*(a) \geq 0.$$

Hence, regret measures the expected cumulative loss in reward caused by not knowing the optimal arm in advance.

A bandit algorithm is said to have *sublinear regret* if

$$\lim_{t \rightarrow \infty} \frac{\text{Regret}_t}{t} = 0, \quad (1.2)$$

or equivalently,

$$\text{Regret}_t = o(t).$$

Using (1.1),

$$\begin{aligned} \frac{\text{Regret}_t}{t} &= \frac{1}{t} \mathbb{E} \left[\sum_{k=1}^t (R^* - R_k) \right] \\ &= R^* - \mathbb{E} \left[\frac{1}{t} \sum_{k=1}^t R_k \right]. \end{aligned} \quad (1.3)$$

Therefore, by (1.2) and (1.3), sublinear regret implies

$$\lim_{t \rightarrow \infty} \mathbb{E} \left[\frac{1}{t} \sum_{k=1}^t R_k \right] = R^*.$$

Thus, the significance of sublinear regret is that the average expected reward of the learning algorithm converges to the best achievable expected reward. Although the algorithm incurs regret while exploring and learning which arm is optimal, the regret per time step vanishes asymptotically.

Question 2

The definition of sublinear regret requires

$$\lim_{T \rightarrow \infty} \frac{\text{Regret}_T}{T} = 0,$$

which is the standard asymptotic criterion used in multi-armed bandit regret analysis. Therefore, a natural numerical test is to estimate the regret at a sequence of increasingly large horizons and examine its asymptotic growth.

Suppose the experiment is repeated independently for M runs, and let $A_t^{(r)}$ denote the arm selected at time t in run r . Since the synthetic bandit environment is known to us, the optimal expected reward is

$$R^* = \max_{a \in \mathcal{A}} q^*(a).$$

The true action values $q^*(a)$ are used only for performance evaluation after an action has been selected; they are not made available to the learning algorithm.

For run r , define the cumulative pseudo-regret at horizon T as

$$C_T^{(r)} = \sum_{t=1}^T \left[R^* - q^*(A_t^{(r)}) \right].$$

Averaging over the M independent reruns gives the empirical estimate

$$\widehat{\text{Regret}}_T = \frac{1}{M} \sum_{r=1}^M C_T^{(r)}.$$

The first diagnostic is the normalised empirical regret

$$\hat{\rho}_T = \frac{\widehat{\text{Regret}}_T}{T}. \quad (2.1)$$

By the definition of sublinear regret, (2.1) should approach zero as T increases. However, examining this quantity at only one finite horizon can be misleading. For example, if

$$\text{Regret}_T = C + cT, \quad c > 0,$$

then

$$\frac{\text{Regret}_T}{T} = \frac{C}{T} + c$$

may decrease substantially with T , even though it eventually converges to the positive constant c and the regret is therefore linear.

To distinguish such transient behaviour from genuinely sublinear growth, we additionally evaluate the incremental regret rate between successive horizons $T_i < T_{i+1}$:

$$\hat{s}_i = \frac{\widehat{\text{Regret}}_{T_{i+1}} - \widehat{\text{Regret}}_{T_i}}{T_{i+1} - T_i}. \quad (2.2)$$

This is the average slope of the cumulative-regret curve over the interval $[T_i, T_{i+1}]$.

The use of (2.2) follows directly from the definition of sublinear regret. Consider geometrically increasing horizons $T_{i+1} = cT_i$, where $c > 1$. Since cumulative pseudo-regret is non-negative and non-decreasing,

$$0 \leq \hat{s}_i \leq \frac{\widehat{\text{Regret}}_{cT_i}}{(c-1)T_i} = \frac{c}{c-1} \frac{\widehat{\text{Regret}}_{cT_i}}{cT_i}.$$

Hence, if the underlying regret is sublinear, the right-hand side approaches zero as the horizon increases, and therefore the incremental regret rate should also approach zero. In contrast, if

$$\text{Regret}_T \sim c_0 T, \quad c_0 > 0,$$

then

$$\hat{s}_i \longrightarrow c_0,$$

so a positive late-horizon plateau in the incremental regret rate provides numerical evidence of linear regret.

As a supplementary diagnostic, the local effective growth exponent is estimated by

$$\hat{\alpha}_i = \frac{\log \widehat{\text{Regret}}_{T_{i+1}} - \log \widehat{\text{Regret}}_{T_i}}{\log T_{i+1} - \log T_i}.$$

If locally $\widehat{\text{Regret}}_T \approx CT^\alpha$, then $\hat{\alpha}_i \approx \alpha$. Thus, $\hat{\alpha}_i$ approaching 1 supports approximately linear growth, whereas values remaining below 1 support sublinear power-law growth.

The proposed numerical procedure is therefore to simulate each algorithm up to a maximum horizon and record its cumulative pseudo-regret at several geometrically increasing horizons, for example

$$T \in \{10^3, 3 \times 10^3, 10^4, 3 \times 10^4, 10^5\}.$$

The experiment is repeated for many independent runs and $\widehat{\text{Regret}}_T$, $\hat{\rho}_T$, $\hat{s}_i$, and $\hat{\alpha}_i$ are calculated.

Numerical behaviour is considered consistent with sublinear regret when, over the largest simulated horizons,

$$\hat{\rho}_T \longrightarrow 0 \quad \text{and} \quad \hat{s}_i \longrightarrow 0,$$

with the effective exponent remaining below 1. Conversely, a positive late-horizon incremental regret rate together with an effective exponent approaching 1 is evidence of linear regret.

Because only finitely many horizons and reruns can be simulated, this test cannot prove an asymptotic regret property. It provides numerical evidence by examining whether the observed large-horizon growth is consistent with the mathematical definition of sublinear regret.

The proposed numerical procedure is summarised in Algorithm 1.

Algorithm 1 Multi-horizon numerical test for sublinear regret

Require: Bandit policy π , horizons $T_1 < \dots < T_L$, number of reruns M

- 1: **for** $r = 1, \dots, M$ **do**
- 2: Simulate π up to T_L
- 3: Record cumulative pseudo-regret $C_{T_i}^{(r)}$ at each horizon T_i
- 4: **end for**
- 5: **for** $i = 1, \dots, L$ **do**
- 6: Estimate $\widehat{\text{Regret}}_{T_i} \leftarrow \frac{1}{M} \sum_{r=1}^M C_{T_i}^{(r)}$
- 7: Compute $\widehat{\rho}_{T_i} \leftarrow \widehat{\text{Regret}}_{T_i} / T_i$
- 8: **end for**
- 9: **for** $i = 1, \dots, L - 1$ **do**
- 10: Compute incremental regret rate

$$\widehat{s}_i \leftarrow \frac{\widehat{\text{Regret}}_{T_{i+1}} - \widehat{\text{Regret}}_{T_i}}{T_{i+1} - T_i}$$

- 11: Compute effective exponent

$$\widehat{\alpha}_i \leftarrow \frac{\log \widehat{\text{Regret}}_{T_{i+1}} - \log \widehat{\text{Regret}}_{T_i}}{\log T_{i+1} - \log T_i}$$

- 12: **end for**
 - 13: Examine the late-horizon trends of $\widehat{\rho}_{T_i}$, $\widehat{s}_i$, and $\widehat{\alpha}_i$
 - 14: **Sublinear evidence:** $\widehat{\rho}_{T_i} \rightarrow 0$, $\widehat{s}_i \rightarrow 0$, and $\widehat{\alpha}_i < 1$
 - 15: **Linear evidence:** $\widehat{s}_i \rightarrow c > 0$ and $\widehat{\alpha}_i \rightarrow 1$
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Question 3

The supplied ϵ -greedy implementation was extended using the multi-horizon numerical procedure proposed in Algorithm 1. For each selected action A_t , the instantaneous pseudo-regret was evaluated as

$$R^* - q^*(A_t),$$

where q^* was used only for post-action performance evaluation and was not available to the learning algorithm.

The six exploration schedules considered were

$$\begin{aligned} \epsilon_t &= 0, & \epsilon_t &= 0.01, & \epsilon_t &= 0.10, \\ \epsilon_t &= \min \left\{ \left(\frac{k \log(t+1)}{t+1} \right)^{1/3}, 1 \right\}, \\ \epsilon_t &= 0.99^t, & \epsilon_t &= 0.90^t. \end{aligned}$$

The $t+1$ form in the polynomial schedule accounts for Python's zero-based indexing. The final exponential-decay case in the supplied code was also corrected to use 0.90^t , consistent with its stated label.

To reduce the risk of mistaking finite-horizon transient behaviour for an asymptotic trend, each policy was simulated for $M = 500$ independent reruns up to

$$T_{\max} = 10^5,$$

with cumulative pseudo-regret recorded at

$$T_i \in \{10^3, 3 \times 10^3, 10^4, 3 \times 10^4, 10^5\}.$$

At these horizons, the normalised regret $\hat{\rho}_{T_i}$, incremental regret rate $\hat{s}_i$, and effective regret exponent $\hat{\alpha}_i$ defined in Question 2 were evaluated.

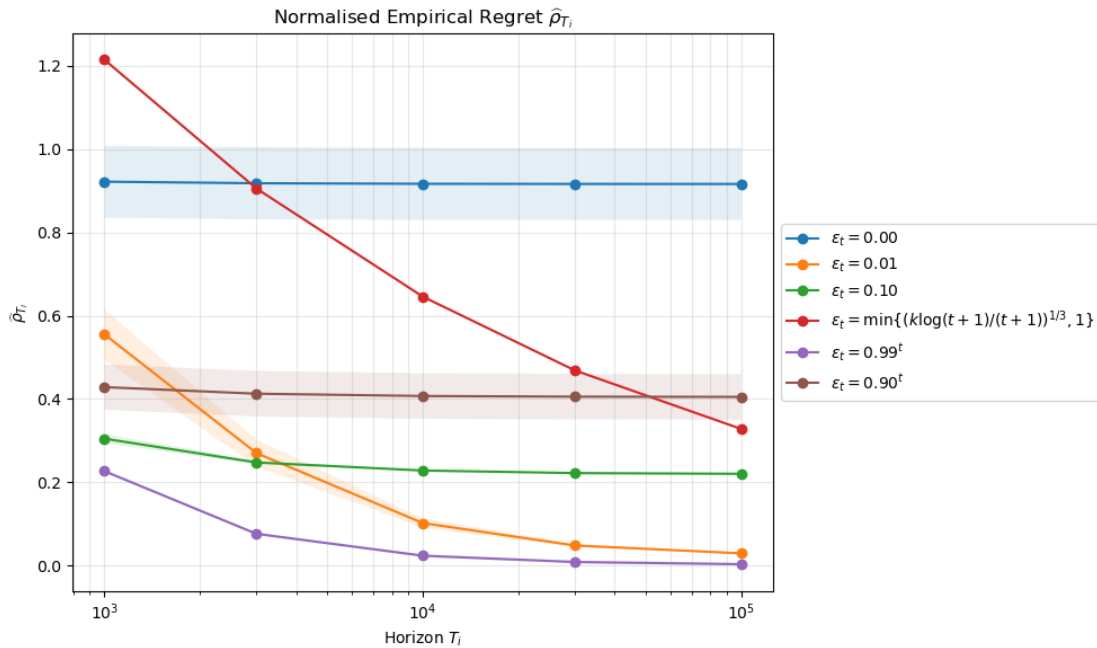


Figure 1: Normalised empirical cumulative pseudo-regret $\hat{\rho}_{T_i}$ for the six ϵ -greedy schedules. Shaded regions represent approximate 95% Monte Carlo confidence intervals.

Figure 1 provides the diagnostic most directly related to the definition of sublinear regret. The curves for $\epsilon_t = 0$, $\epsilon_t = 0.10$, and $\epsilon_t = 0.90^t$ clearly approach positive values and are therefore inconsistent with $\hat{\rho}_{T_i} \rightarrow 0$. The cases $\epsilon_t = 0.01$ and $\epsilon_t = 0.99^t$ are less obvious from normalised regret alone because their curves continue decreasing over the simulated horizons. This motivates the additional diagnostics introduced in Question 2.

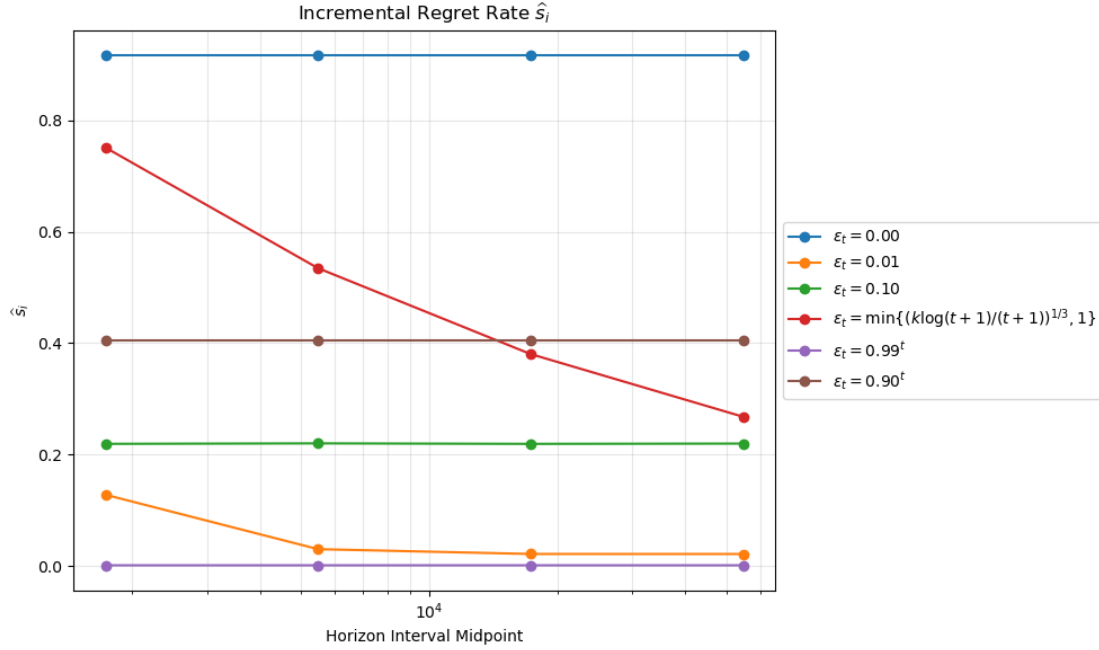


Figure 2: Incremental regret rate $\hat{s}_i$ between consecutive simulation horizons.

Figure 2 shows whether additional regret is still accumulating at a non-zero rate. For a linear-regret policy, $\hat{s}_i$ should approach a positive constant, whereas for sublinear regret it should decrease towards zero. The polynomial schedule is the only case exhibiting persistent decay over all four intervals:

$$0.7507 \rightarrow 0.5345 \rightarrow 0.3805 \rightarrow 0.2681.$$

In contrast, the other schedules eventually exhibit an approximately constant positive incremental regret rate.

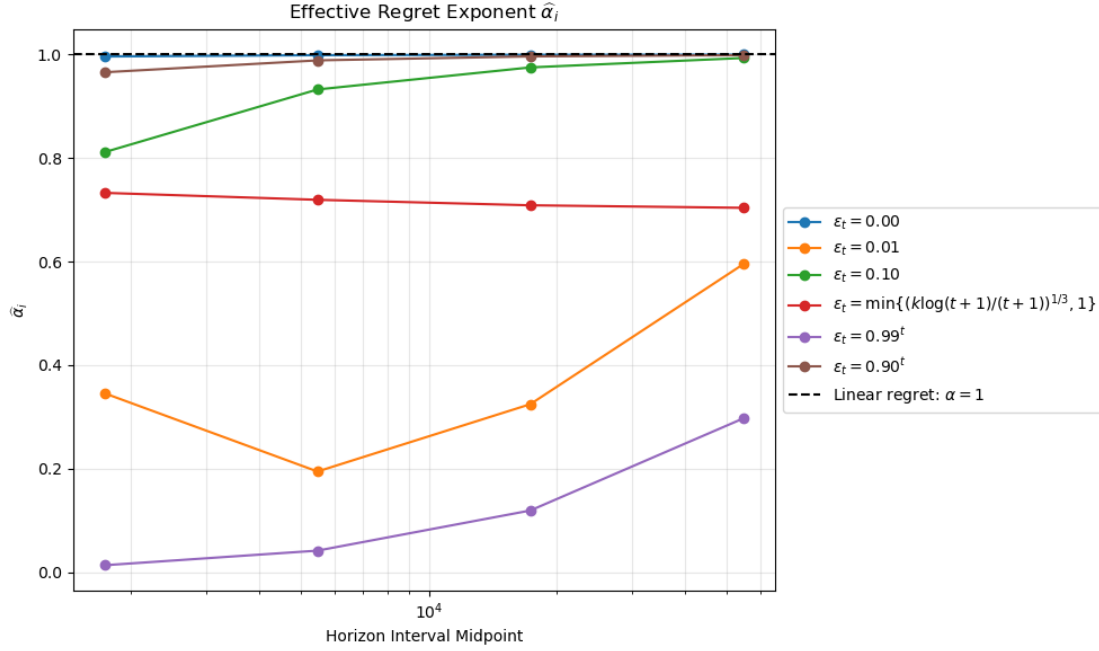


Figure 3: Effective regret exponent $\hat{\alpha}_i$ between consecutive horizons. The dashed line $\alpha = 1$ corresponds to linear power-law growth.

The effective exponents in Figure 3 provide a supplementary indication of the growth order. In particular, $\hat{\alpha}_i$ approaches one for several of the policies exhibiting linear regret, whereas the polynomial schedule remains well below one, decreasing from approximately 0.733 to 0.704.

The complete multi-horizon diagnostics are summarised in Table 1.

Table 1: Multi-horizon numerical diagnostics for the six ϵ -greedy schedules.

Schedule	$\hat{\rho}_{10^5}$	$\hat{s}_1 \rightarrow \hat{s}_4$	$\hat{\alpha}_1 \rightarrow \hat{\alpha}_4$	Assessment
$\epsilon_t = 0$	0.9158	0.9157 $\rightarrow$ 0.9157 $\rightarrow$ 0.9157 $\rightarrow$ 0.9157	0.9963 $\rightarrow$ 0.9988 $\rightarrow$ 0.9996 $\rightarrow$ 0.9999	Linear
$\epsilon_t = 0.01$	0.0300	0.1282 $\rightarrow$ 0.0306 $\rightarrow$ 0.0220 $\rightarrow$ 0.0219	0.3456 $\rightarrow$ 0.1946 $\rightarrow$ 0.3246 $\rightarrow$ 0.5948	Linear
$\epsilon_t = 0.10$	0.2207	0.2194 $\rightarrow$ 0.2203 $\rightarrow$ 0.2194 $\rightarrow$ 0.2200	0.8115 $\rightarrow$ 0.9326 $\rightarrow$ 0.9752 $\rightarrow$ 0.9934	Linear
Polynomial decay	0.3283	0.7507 $\rightarrow$ 0.5345 $\rightarrow$ 0.3805 $\rightarrow$ 0.2681	0.7329 $\rightarrow$ 0.7195 $\rightarrow$ 0.7089 $\rightarrow$ 0.7040	Sublinear
$\epsilon_t = 0.99^t$	0.0040	0.001699 $\rightarrow$ 0.001697 $\rightarrow$ 0.001697 $\rightarrow$ 0.001697	0.0135 $\rightarrow$ 0.0417 $\rightarrow$ 0.1193 $\rightarrow$ 0.2970	Emerging linear
$\epsilon_t = 0.90^t$	0.4052	0.4049 $\rightarrow$ 0.4049 $\rightarrow$ 0.4049 $\rightarrow$ 0.4049	0.9658 $\rightarrow$ 0.9888 $\rightarrow$ 0.9965 $\rightarrow$ 0.9989	Linear

For $\epsilon_t = 0$, no systematic exploration occurs. The algorithm may therefore commit permanently to a suboptimal arm. Its incremental regret rate remains at approximately 0.916, while $\hat{\alpha}_i \rightarrow 1$, providing clear evidence of linear regret.

For the fixed exploration rate $\epsilon_t = 0.10$, $\hat{s}_i$ remains almost exactly constant at approximately 0.22 and $\hat{\alpha}_i$ approaches one. Hence, its cumulative regret is clearly linear. The case $\epsilon_t = 0.01$ illustrates why the incremental-rate diagnostic is useful: although its normalised regret is still decreasing at $T = 10^5$, its incremental regret rate settles at approximately 0.0219. Thus the observed behaviour is consistent with a regret curve of the form

$$\text{Regret}_T \approx C + cT, \quad c > 0,$$

where the transient term C/T causes the normalised regret to continue decreasing before eventually approaching the positive constant c .

The polynomial schedule

$$\epsilon_t = \min \left\{ \left(\frac{k \log(t+1)}{t+1} \right)^{1/3}, 1 \right\}$$

is the only case that is numerically consistent with sublinear regret. Its normalised regret continues to decrease, its incremental regret rate decreases persistently towards zero, and its effective exponent remains well below one across the largest simulated horizons.

For $\epsilon_t = 0.90^t$, exploration decays so rapidly that the algorithm can become permanently committed to a suboptimal arm. Correspondingly, $\hat{s}_i$ approaches the positive value 0.4049 and $\hat{\alpha}_i \rightarrow 1$, indicating linear regret.

The slower exponential schedule $\epsilon_t = 0.99^t$ performs substantially better at finite horizons, but the multi-horizon test reveals behaviour that was not apparent in the shorter experiment. Its incremental regret rate settles at the small but positive value

$$\hat{s}_i \approx 0.001697,$$

while its effective exponent increases from approximately 0.014 to 0.297 as the horizon grows. This is numerically consistent with an emerging linear-regret component rather than genuine sublinear convergence.

Therefore, among the six tested exploration schedules, the only policy numerically consistent with sublinear regret is

$$\epsilon_t = \min \left\{ \left(\frac{k \log(t+1)}{t+1} \right)^{1/3}, 1 \right\}.$$

The numerical conclusion agrees with the theoretical result that exploration must decay sufficiently slowly to maintain learning while its long-run frequency approaches zero. Since only finite horizons and finitely many reruns are simulated, these results provide numerical evidence of the asymptotic growth regime rather than a mathematical proof.

Question 4

To investigate whether UCB learns the true value of each arm, the algorithm was simulated for $T = 10^4$ steps over $M = 300$ independent reruns for $c \in \{0, 1, 2\}$. Each arm was selected once initially to avoid an undefined confidence bonus. Thereafter, actions were selected using the UCB rule and the empirical action-value estimate $Q_t(a)$ was updated by the sample average.

Since the true values $q^*(a)$ are known in the simulated environment, the estimation accuracy of each arm was measured using

$$\text{RMSE}_t(a) = \sqrt{\frac{1}{M} \sum_{r=1}^M \left(Q_t^{(r)}(a) - q^*(a) \right)^2}.$$

A decreasing RMSE therefore indicates that the empirical estimate of an arm is approaching its true value.

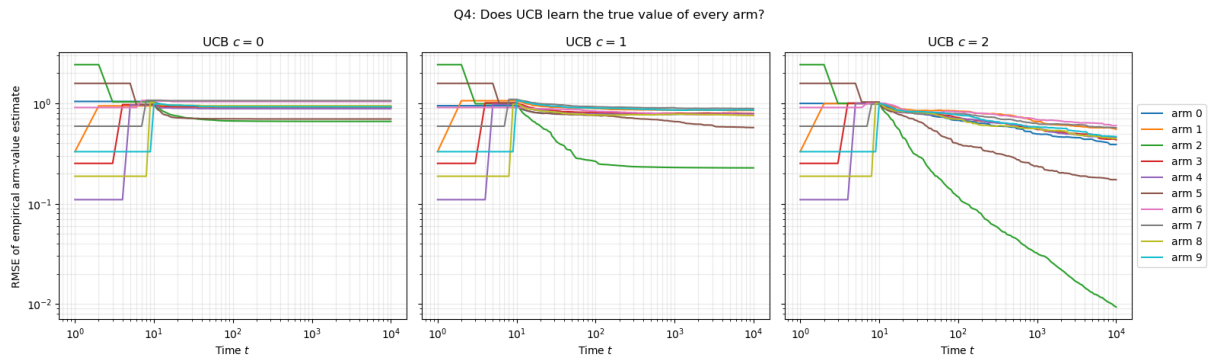


Figure 4: RMSE of the empirical value estimate for each arm under UCB with $c = 0, 1, 2$, averaged over 300 independent reruns.

As shown in Figure 4, the amount of exploration has a strong effect on value estimation. For $c = 0$, the confidence bonus vanishes and the algorithm becomes purely greedy after the initial pulls. Consequently, many arms are never revisited and their estimation errors do not decrease. The final mean arm-wise RMSE is approximately 0.898.

For $c = 1$, UCB continues to revisit uncertain arms, reducing the final mean arm-wise RMSE to approximately 0.725. However, highly suboptimal arms are sampled only a few times by $T = 10^4$, so their estimates remain noisy. Increasing the exploration coefficient to $c = 2$ produces substantially more accurate estimates across the arms, reducing the final mean arm-wise RMSE to approximately 0.409.

Hence, the experiment indicates that exploratory UCB ($c > 0$) progressively learns the true arm values, but convergence is much faster for frequently selected competitive arms than for highly suboptimal arms. The degenerate case $c = 0$ does not in general learn every arm because it ceases systematic exploration.

Question 5

The UCB and ϵ -greedy algorithms were compared on the same 10-armed bandit over a horizon of $T = 10^4$, with the experiment repeated over $M = 300$ independent reruns. For each run, cumulative pseudo-regret was computed as

$$\sum_{t=1}^T [R^* - q^*(A_t)],$$

and then averaged across reruns. The comparison includes UCB with $c \in \{0, 1, 2\}$ and the six ϵ -greedy schedules considered previously.

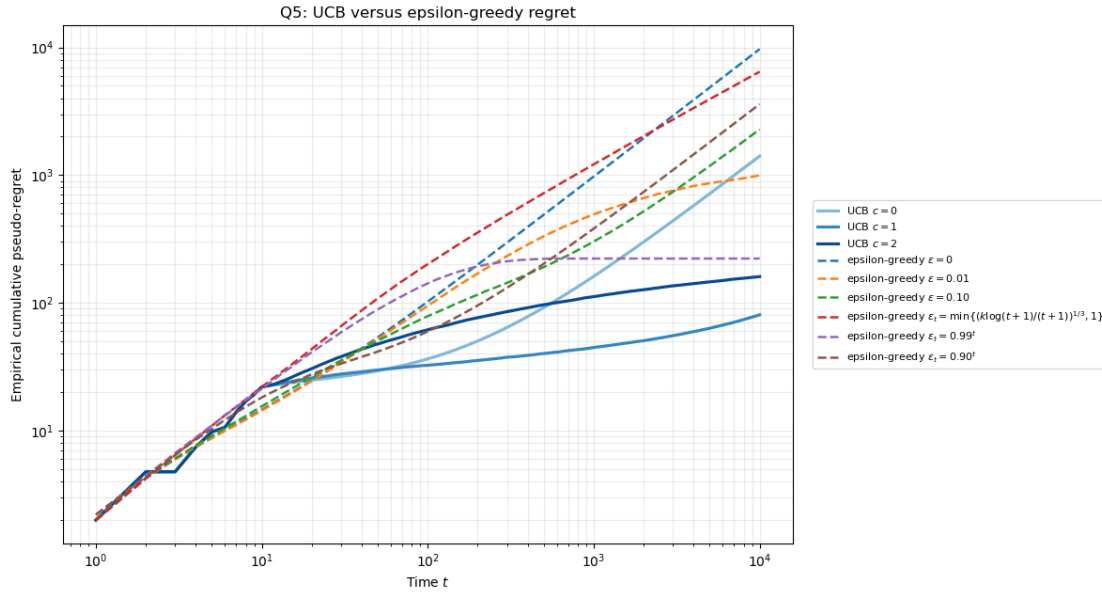


Figure 5: Empirical cumulative pseudo-regret of UCB and ϵ -greedy algorithms over $T = 10^4$ time steps.

Figure 5 shows that the exploratory UCB algorithms achieve substantially lower finite-horizon regret than the tested ϵ -greedy policies. At $T = 10^4$, the empirical cumulative pseudo-regret is approximately

$$80.6 \quad \text{for UCB } c = 1, \quad 160.3 \quad \text{for UCB } c = 2,$$

whereas the best-performing ϵ -greedy schedule, $\epsilon_t = 0.99^t$, has regret approximately 222.0.

Among the UCB configurations, $c = 1$ gives the lowest regret in this experiment. Increasing the exploration coefficient to $c = 2$ incurs a larger exploration cost and therefore slightly higher regret, although the results of Question 4 show that this additional exploration produces more accurate estimates of the individual arm values. In contrast, $c = 0$ does not perform systematic exploration and incurs substantially larger regret (≈ 1408).

The fixed ϵ -greedy policies also accumulate considerably more regret: at $T = 10^4$, the regret is approximately 994 for $\epsilon = 0.01$ and 2278 for $\epsilon = 0.10$. Persistent random exploration continues to select suboptimal arms even after the optimal arm has been identified. The polynomially decaying schedule has even larger finite-horizon regret (≈ 6445), because it continues to explore aggressively at this horizon, despite its sublinear asymptotic behaviour identified in Question 3.

Therefore, for this bandit instance and finite horizon, UCB provides the best exploration–exploitation trade-off. Unlike ϵ -greedy, which explores randomly, UCB directs exploration toward

arms with greater uncertainty. This allows the exploratory UCB algorithms to identify the optimal arm while avoiding much of the unnecessary exploration cost incurred by the tested ϵ -greedy policies.

Question 6

The gradient-bandit and UCB algorithms were compared on the same 10-armed bandit over $T = 10^4$ time steps and $M = 300$ independent reruns. For each run, cumulative pseudo-regret was computed as

$$\sum_{t=1}^T [R^* - q^*(A_t)],$$

and averaged across reruns. The comparison uses UCB with $c \in \{0, 1, 2\}$ and the gradient-bandit algorithm with $\alpha \in \{0, 0.1, 0.5\}$.

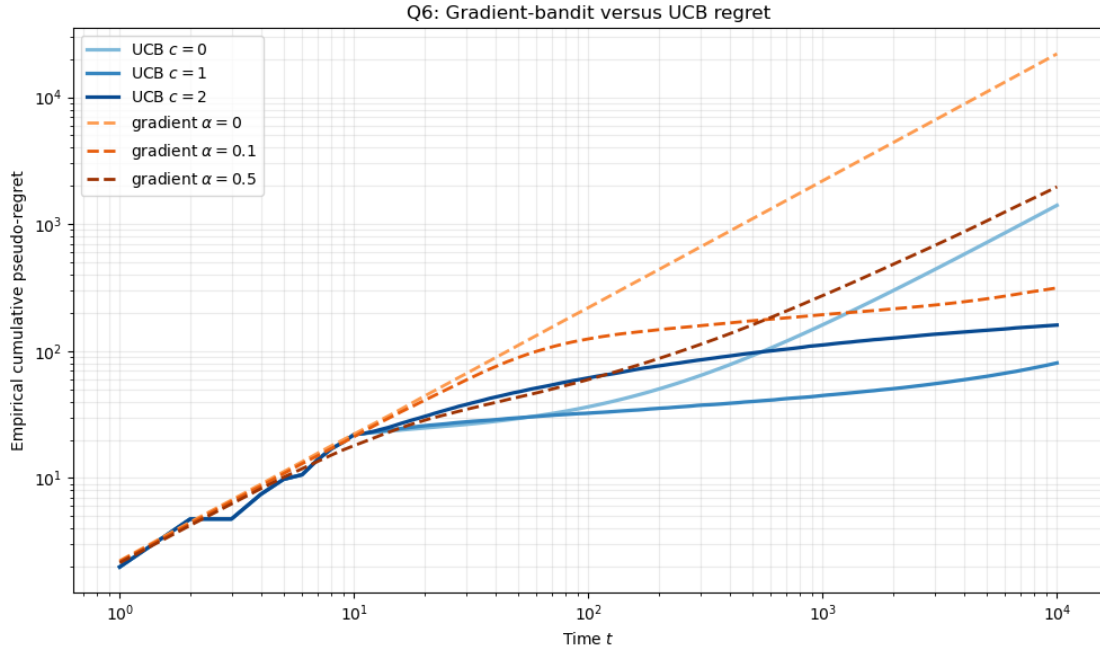


Figure 6: Empirical cumulative pseudo-regret of the gradient-bandit and UCB algorithms over $T = 10^4$ time steps.

The cumulative pseudo-regret at $T = 10^4$ is summarised in Table 2.

Table 2: Empirical cumulative pseudo-regret at $T = 10^4$.

Algorithm	Cumulative pseudo-regret
UCB, $c = 1$	80.58
UCB, $c = 2$	160.29
Gradient, $\alpha = 0.1$	314.58
UCB, $c = 0$	1408.07
Gradient, $\alpha = 0.5$	1968.16
Gradient, $\alpha = 0$	21978.00

As shown in Figure 6, the exploratory UCB configurations achieve the lowest regret among the tested algorithms. In particular, UCB with $c = 1$ performs best, with a cumulative pseudo-regret of approximately 80.6, followed by UCB with $c = 2$ at approximately 160.3.

Among the tested gradient-bandit configurations, $\alpha = 0.1$ performs best, with cumulative pseudo-regret of approximately 314.6. This is substantially better than the non-exploratory UCB case

$c = 0$, but is still approximately four times the regret obtained by UCB with $c = 1$. Increasing the gradient learning rate to $\alpha = 0.5$ gives substantially larger regret, indicating that the more aggressive preference updates are more sensitive to stochastic reward fluctuations. With $\alpha = 0$, the preferences remain unchanged and the softmax policy remains uniform, so the algorithm does not learn and incurs approximately linear regret.

Hence, for this stationary bandit instance and the tested parameter values, the exploratory UCB algorithms provide a more regret-efficient exploration–exploitation trade-off than the gradient-bandit configurations.

Question 7

To investigate the sensitivity of the gradient-bandit algorithm to the learning rate, the experiment was repeated for

$$\alpha \in \{0.01, 0.05, 0.1, 0.5, 1, 2, 5\}.$$

For each value of α , the algorithm was run over $T = 10^4$ time steps and multiple independent reruns. At each time step, the probability assigned by the softmax policy to the true optimal arm

$$a^* \in \arg \max_{a \in \mathcal{A}} q^*(a)$$

was recorded and averaged across reruns. Thus, convergence to the optimal decision corresponds to the mean probability of selecting a^* approaching one.

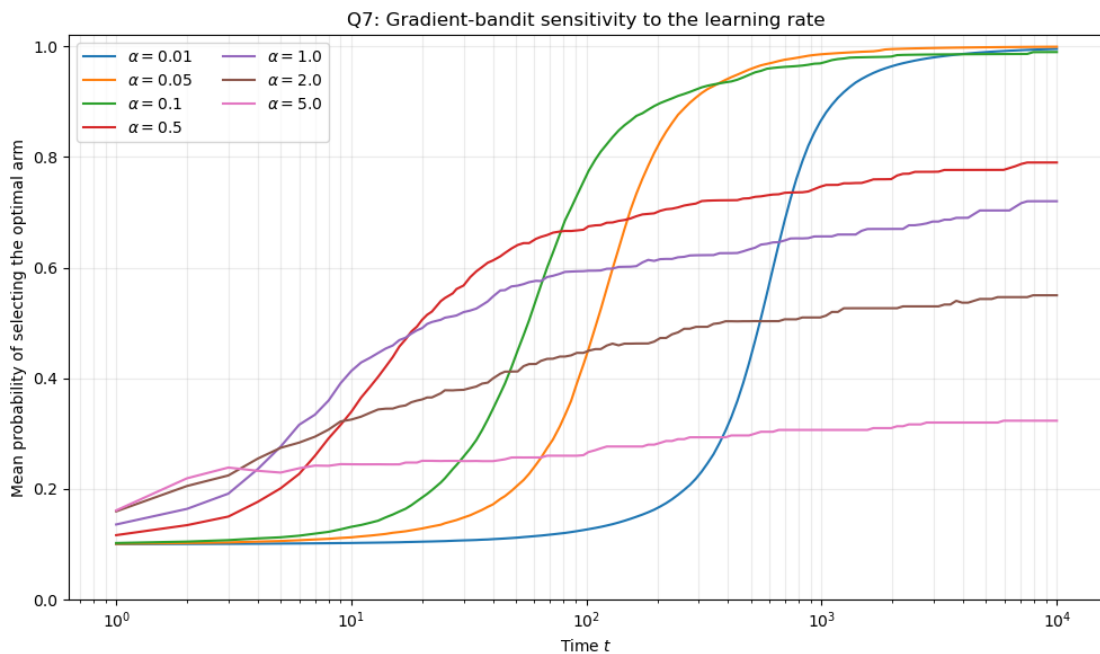


Figure 7: Mean probability of selecting the optimal arm for different gradient-bandit learning rates.

Figure 7 shows that the gradient-bandit algorithm does *not* exhibit convergence to the optimal arm for every choice of $\alpha > 0$. For the smaller learning rates $\alpha = 0.01, 0.05$, and 0.1 , the probability of selecting the optimal arm approaches approximately one by the end of the simulation. However, performance deteriorates as the learning rate becomes large. At $T = 10^4$, the optimal-arm probabilities are only approximately $0.79, 0.72, 0.55$, and 0.33 for $\alpha = 0.5, 1, 2$, and 5 , respectively.

The learning rate controls the magnitude of each stochastic preference update. A small α produces slower but more stable learning, whereas a large α causes individual noisy reward observations to produce large changes in the action preferences. Consequently, the policy continues to fluctuate rather than concentrating reliably on the optimal arm.

Therefore, the numerical experiment provides evidence that convergence is sensitive to the choice of learning rate: sufficiently small values of α give behaviour consistent with convergence, whereas large constant learning rates do not. Since the experiment has a finite horizon, it does not constitute an asymptotic proof, but it is sufficient to show that $\alpha > 0$ alone does not ensure satisfactory convergence in practice.